WORKSHEET 7.1

Writing a Differential Equation from a Description

Unit 7: Differential Equations

Key ideas

- "The rate of change of y" is a derivative: $\dfrac{dy}{dt}$.
- "Proportional to" means a constant times: ky. "Inversely proportional to" means a constant divided by: $\dfrac{k}{y}$.
- When something is added at one rate and removed at another, the equation is (rate in) − (rate out).
- A given rate at a given value fixes the constant k. A decreasing quantity has a negative rate.

Worked examples

Example 1 · A direct translation

The rate of change of y with respect to t is proportional to the square of y. Write a differential equation.

Solution. Rate: $\dfrac{dy}{dt}$. Proportional to y^2: ky^2.

$$\frac{dy}{dt} = ky^2.$$

Example 2 · In and out

A savings account with balance B dollars earns interest at 3% per year, and 2000 dollars per year is withdrawn. Write a differential equation, and find the balance that stays constant.

Solution. Rate in $0.03B$; rate out 2000.

$$\frac{dB}{dt} = 0.03B - 2000\text{; constant when } B \approx 66\,667 \text{ dollars.}$$

Example 3 · Finding the constant

A frozen meal at temperature T warms in a room at 21 °C at a rate proportional to $21 - T$. When $T = 5$ it is warming at 1.6 °C per minute. Write the differential equation.

Solution. $\dfrac{dT}{dt} = k(21 - T)$, and $1.6 = k(16)$.

$$\frac{dT}{dt} = 0.1(21 - T).$$

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Practice

CORE

1. Write a differential equation: the rate of change of y with respect to x is proportional to x.

2. Write a differential equation: the rate of change of P with respect to t is inversely proportional to P.

3. Write a differential equation: a quantity N decreases at a rate proportional to N. State the sign of your constant.

4. A quantity satisfies $\frac{dy}{dt} = ky$, and $\frac{dy}{dt} = 12$ when $y = 3$. Find k.

EXAM LEVEL

5. A pollutant enters a lake at 12 kg per day, and each day 5% of the amount Q present is removed. Write a differential equation for Q, and find the amount at which Q is steady.

6. A population P grows at a rate proportional to the product of P and $2000 - P$. Write a differential equation, and find the value of P at which the growth is fastest.

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7. The depth h cm of water in a tank falls at a rate proportional to $\sqrt{h}$. When $h = 16$ the depth is falling at 2 cm per minute. Write the differential equation, including the constant.

8. In $\dfrac{dM}{dt} = kM$, the mass M is in grams and the time t in hours. What are the units of k?

Question 9 · Free-response practice

A trainee's score S on a skills test, out of 100, increases with practice at a rate proportional to $100 - S$, where t is the number of weeks of practice. When $S = 40$, the score is increasing at 3 points per week.

(a) Write a differential equation for S in terms of a constant k.

(b) Find the value of k.

(c) Find the rate at which the score is increasing when $S = 80$, and describe what the model predicts as S approaches 100.

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Worked solutions

1. $\dfrac{dy}{dx} = kx$.
2. $\dfrac{dP}{dt} = \dfrac{k}{P}$.
3. $\dfrac{dN}{dt} = -kN$ with $k > 0$ (or $\dfrac{dN}{dt} = kN$ with $k < 0$).
4. $12 = 3k$, so $k = 4$.
5. $\dfrac{dQ}{dt} = 12 - 0.05Q$, steady when $Q = 240$ kg.
6. $\dfrac{dP}{dt} = kP(2000 - P)$; $P(2000 - P)$ is greatest at $P = 1000$.
7. $\dfrac{dh}{dt} = -k\sqrt{h}$ with $-2 = -k(4)$: $\dfrac{dh}{dt} = -0.5\sqrt{h}$.
8. Grams per hour divided by grams: per hour.
9. (a) $\dfrac{dS}{dt} = k(100 - S)$.
   (b) $3 = k(60)$, so $k = 0.05$.
   (c) $0.05(20) = 1$ point per week. As $S \to 100$ the rate tends to 0: improvement slows and the score levels off below 100.

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